

Texture Resonance Retrieval for Guitar-Effect Preset Selection

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Abstract—Digital audio workstations expose editable effect chains, but mapping perceptual intent to valid low-level parameters remains difficult because acoustic similarity does not guarantee parameter transferability. We study editable guitar-effect control as executable-neighborhood retrieval over real presets. We propose Texture Resonance Retrieval (TRR), a texture-aware retrieval prior that represents mid-level Wav2Vec2 activations with second-order Gram statistics. On an audited 204-query benchmark with 1,063 candidate presets, TRR is the strongest full-coverage top-1 retrieval method among evaluated baselines, reducing normalized parameter distance from 0.2002 to 0.1454. Exemplar-preserving projection (EPR-K5) further improves TRR to 0.1334 while retaining neighborhood provenance. A range-projected MLP is competitive on tolerance accuracy but weaker on normalized distance, recall, and cosine similarity, clarifying the value of retrieval over executable exemplars. This framing treats editable provenance as an evaluation target rather than only an interface property under auditable held-out protocols. Mechanism, leakage, hard-split, degraded-input, and listening diagnostics bound the claim to the current guitar-effect benchmark.

Index Terms—guitar-effect preset retrieval, audio retrieval, digital audio workstations, texture representation, interpretable audio systems

I. INTRODUCTION

The field of neural audio synthesis currently faces a fundamental tension between *fidelity* and *control*. Recent high-fidelity generative models, primarily based on latent diffusion [1, 2], can achieve strong perceptual quality but often operate as “black boxes.” They generate finalized waveforms where internal attributes—such as the attack time of a compressor or the decay of a reverb—are entangled and inaccessible for post-hoc editing. Conversely, Differentiable Digital Signal Processing (DDSP) [3] offers interpretable, structured control but often suffers from the ill-posed nature of the inverse parameter estimation problem.

This tension motivates the development of retrieval-based audio control systems that can map user intent to interpretable DSP parameters. By retrieving relevant presets rather than generating parameters from scratch, such systems are designed to produce editable outputs that integrate with DAW workflows. In this view, retrieval is valuable not because it solves the inverse problem in closed form, but because it places the query in a plausible executable neighborhood of parameter space that the user can still inspect and refine.

The key challenge is grounding user descriptions in parameter space via retrieval. Standard audio representations (e.g., CLAP-style embeddings or mean-pooled Wav2Vec2 features) summarize an audio segment into a single vector; while effective for coarse semantics, such pooling can discard temporal co-activation patterns that are critical for texture-dominant effects (e.g., tremolo modulation or signal-dependent distortion).

To bridge these gaps, we study a retrieval-grounded preset-selection setting and focus on its core retrieval component: **Texture Resonance Retrieval (TRR)**. TRR models audio style using Gram matrices of deep feature activations, explicitly capturing the correlation structure that defines temporal texture. We use this representation as a texture-aware retrieval prior that biases search toward executable presets with preserved effect-relevant structure. We evaluate TRR on a curated guitar-effect preset benchmark of 1,063 candidates and 204 held-out queries.

Concretely, this paper makes three evidence-bounded contributions. First, we formulate editable guitar-effect control as parameter-transferability-aware executable neighborhood retrieval, distinguishing it from waveform generation, general semantic retrieval, and direct parameter regression. Second, we evaluate TRR as a texture-aware retrieval prior built on mid-level deep features; we report second-order Gram statistics as an effective aggregation operator and use controlled ablations to clarify that the advantage is coupled with mid-level layer selection. Third, we introduce exemplar-preserving projection (EPR-K5) and report an audited evaluation package—including top-1 retrieval, near-duplicate filtering, parameter-cluster hard split, direct-regression boundary, multimodal degradation diagnostics, and exploratory listening evidence—to bound the claim to the current benchmark.

II. RELATED WORK

Our work lies at the intersection of intelligent music production, neural audio synthesis, audio representation learning, and retrieval-grounded parameter control.

Semantic control of audio effects has long been studied in intelligent music production, including semantic-to-parameter mapping in DAW settings [4], interactive control learning [5], and recent multimodal synthesizer retrieval tools [6]. These systems motivate editable parameter control, but they do not directly address whether retrieved presets form parameter-transferable neighborhoods under auditable held-out protocols. The SAFE project [4] established semantic audio descriptors for retrieval, yet its descriptors are human-annotated first-order labels rather than learned second-order texture statistics, and its evaluation does not report parameter-space transferability on held-out splits. Recent automixing and parameter-optimization systems [7–9] optimize effect chains via gradient descent or Bayesian methods, yielding strong numerical results but producing dense parameter vectors without exemplar provenance. We position TRR as a retrieval-based alternative that guarantees executability and preserves editability by anchoring outputs to real preset exemplars.

Neural audio synthesis and differentiable DSP provide complementary foundations. Text-to-audio and latent-diffusion

systems [1, 2, 10] generate high-quality waveforms, while DDSP-style methods [3, 11, 12] and neural emulation of analog audio circuits [13] expose more structured controls. Recent work further pursues large-scale synthetic augmentation and contrastive learning for guitar-effects encoders [14], though evaluation has focused on classification accuracy rather than parameter-space retrieval on held-out presets. Recent intelligent music production systems also pursue direct parameter control: LLM2Fx [15] predicts audio effect parameters from natural language, and ST-ITO [9] optimizes effect parameters via inference-time gradient descent. These approaches regress or optimize parameters without exemplar provenance. Human-computer interaction studies with practising musicians confirm that editability, provenance, and DAW integration are decisive adoption factors for creative AI tools [16–18]. Our target is different: rather than synthesizing final audio or regressing parameters without provenance, we retrieve executable presets that remain inspectable and editable in downstream DAW workflows. Retrieval trades some optimality for guaranteed feasibility and human-readable provenance.

Modern audio encoders such as CLAP [19], PaSST [20], PANNs [21], and Wav2Vec2 [22] provide strong first-order or semantic embeddings. However, perceptual similarity does not necessarily imply parameter transferability for effect-chain control. Inspired by Gram-matrix style representations [23, 24] and audio texture analysis [25, 26], TRR uses second-order deep-feature statistics as a retrieval descriptor for executable preset selection. This connects to a broader literature on audio texture perception and synthesis, where stationary statistical properties of the auditory periphery have been shown to characterize texture identity [27]. In contrast to document-centric retrieval-augmented generation [28] or open-ended music generation, the retrieval target here is a finite set of valid parameter exemplars, and success is measured in parameter space with explicit coverage, leakage, and validity audits.

III. PROBLEM FORMULATION

We formalize audio-effect preset selection as constrained retrieval over executable parameters. Let $\Phi : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$ denote a DSP rendering operator and let $\Theta \subset \mathbb{R}^d$ denote the parameter space of a multi-effect guitar processing chain. The feasible set is bounded and structurally constrained:

$$\Theta = \{\theta \in \mathbb{R}^d \mid \mathbf{1} \preceq \theta \preceq \mathbf{u}, \quad \mathcal{C}(\theta) = 1\} \quad (1)$$

where $\mathcal{C}$ encodes plugin-specific validity constraints. Given a query $q = (t, \mathbf{a}_{\text{ref}})$, containing a text description and optionally an audio reference, the objective is to return a valid, editable parameter vector:

$$\tilde{\theta} \in \Theta \quad \text{given } q, \quad (2)$$

where query consistency is operationalized through parameter-space evaluation against held-out presets. Because the inverse mapping from user intent to parameters is ill-posed and the DSP validity constraints are strict, we retrieve a top- K executable neighborhood from a finite preset database:

$$\theta_{\text{selected}} = \text{Top-}K(q) \quad (3)$$

Retrieval supplies a mode-seeking step that places the query in a plausible neighborhood of Θ ; projection and validity repair then operate only within this retrieved support. This framing distinguishes our task from unconstrained waveform generation and from direct dense parameter regression without exemplar provenance.

IV. METHODOLOGY

We implement the retrieval-based framework defined in Section III via **Texture Resonance Retrieval (TRR)** to retrieve candidate presets using texture statistics derived from mid-level deep features.

A. System Architecture

The system realizes $\Phi(\mathbf{x}_{\text{in}}; \theta)$ through a decoupled design over a switchable nine-module topology: Compressor $\rightarrow$ Driver $\rightarrow$ Screamer $\rightarrow$ Equaliser $\rightarrow$ Chorus $\rightarrow$ Flanger $\rightarrow$ Phaser $\rightarrow$ Delay $\rightarrow$ Reverb.

The architecture has two components. The real-time DSP engine executes validated parameters as a deterministic audio graph. The retrieval module runs asynchronously from the audio callback, maps q to candidate presets, and returns parameters only after validity checking. This separation keeps the submitted method focused on retrieval quality rather than claiming unverified real-time inference.

B. Texture Resonance Retrieval (TRR)

The core challenge in texture-aware retrieval lies in computing a similarity metric for the audio modality that preserves effect-relevant dynamics. Standard embeddings (e.g., mean-pooled Wav2Vec2 features) collapse temporal structure into a static vector, losing information that is critical for effects like modulation and distortion.

To address this, we introduce **Texture Resonance Retrieval (TRR)**. Drawing on the hypothesis that audio texture is encoded in second-order feature statistics [27], we compute the Gram matrix of deep activations. In our formulation, *texture* refers to stationary statistical properties of the audio signal that are invariant to exact temporal alignment. This focus on stationary co-activation structure is what distinguishes TRR from first-order semantic embeddings that summarize an entire clip into a single vector.

Let $\mathbf{H}_\ell \in \mathbb{R}^{T_{\text{frm}} \times 768}$ be the frame-level activation map from layer ℓ of a pre-trained Wav2Vec2 Base model. In the benchmarked implementation we use the mid-level layer set $\mathcal{L} = \{4, 5, 6\}$, apply a frozen Xavier-initialized random projection to 64 dimensions, compute a Gram matrix per layer, average the resulting 64×64 matrices, flatten the result into a 4096-dimensional vector, and L2-normalize it. The averaged texture descriptor can be written as

$$\bar{\mathbf{G}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{1}{T_{\text{frm}}} \hat{\mathbf{H}}_\ell^\top \hat{\mathbf{H}}_\ell, \quad (4)$$

where $\hat{\mathbf{H}}_\ell$ denotes the projected feature map. The entry $\bar{\mathbf{G}}_{ij}$ captures the co-activation of projected channels i and

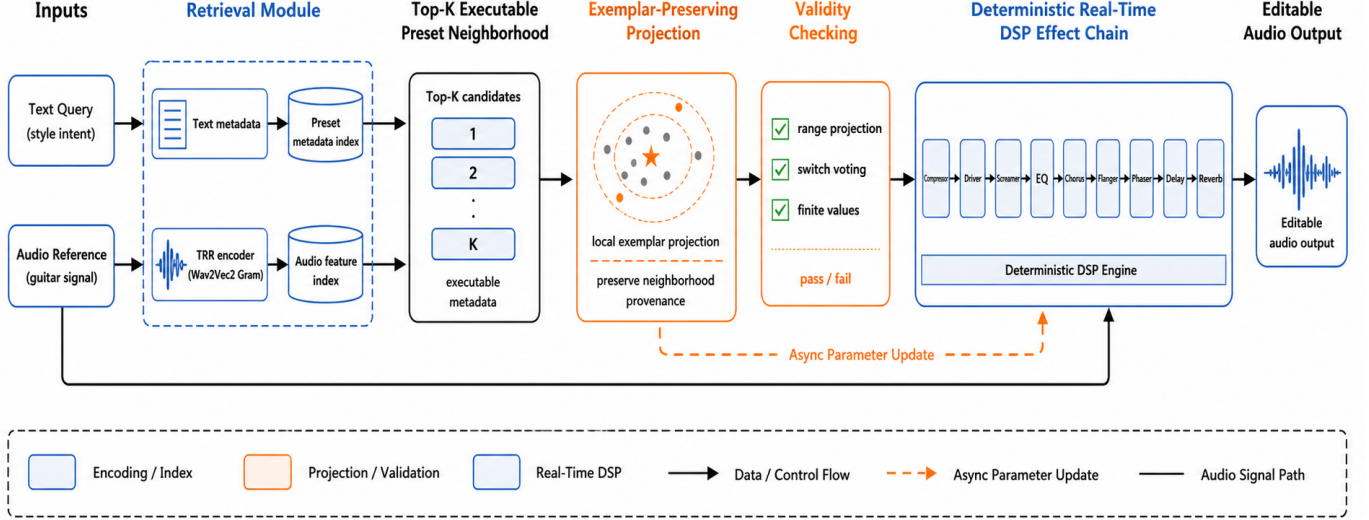


Fig. 1. Overall framework of the retrieval-grounded guitar-effect control system. The figure summarizes the modular flow from query inputs to asynchronous retrieval, validity-checked parameter update, and executable effect-chain output. It is a framework illustration rather than an experimental result figure.

Algorithm 1 TRR Encoding

Require: Audio sample a , Wav2Vec2 model $\mathcal{W}$, layer set $\mathcal{L} = \{4, 5, 6\}$, frozen random linear projection $P : \mathbb{R}^{768} \rightarrow \mathbb{R}^{64}$

Ensure: Texture embedding $\mathbf{g}_{\text{norm}}$

1. Preprocessing: Normalize a and pad to minimum length.

for each $\ell \in \mathcal{L}$ **do**

$\mathbf{H}_\ell \leftarrow \mathcal{W}_{\text{layer } \ell}(a) \in \mathbb{R}^{T_{\text{frm}} \times 768}$

$\hat{\mathbf{H}}_\ell \leftarrow P(\mathbf{H}_\ell) \in \mathbb{R}^{T_{\text{frm}} \times 64}$ {Frozen; Xavier init. used only for initial variance scaling}

$\mathbf{G}_\ell \leftarrow \frac{1}{T_{\text{frm}}} \hat{\mathbf{H}}_\ell^\top \hat{\mathbf{H}}_\ell \in \mathbb{R}^{64 \times 64}$

end for

$\mathbf{G} \leftarrow \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{G}_\ell$

$\mathbf{g}_{\text{norm}} \leftarrow \text{vec}(\mathbf{G}) / \|\text{vec}(\mathbf{G})\|_2$

return $\mathbf{g}_{\text{norm}}$

j irrespective of their absolute temporal position. This time-invariance allows TRR to match textures (e.g., “fast tremolo”) even if the reference and database entries are not phase-aligned. L2 normalization of $\text{vec}(\tilde{\mathbf{G}})$ is equivalent to Frobenius normalization of the matrix representation, and similarity is then computed as cosine similarity between the flattened descriptors. Computationally, Gram-matrix construction scales as $O(D^2 \cdot T_{\text{frm}})$ after projection, where $D = 64$ in the reported implementation. This bound covers only the post-Wav2Vec2 Gram construction; the Wav2Vec2 forward pass is a separate feature-extraction cost and dominates when features are not cached.

TRR is a second-order summary of deep features: it retains cross-channel correlation structure while discarding absolute temporal alignment. For a frame permutation matrix $\mathbf{P}$, $(\mathbf{P}\mathbf{H})^\top(\mathbf{P}\mathbf{H}) = \mathbf{H}^\top\mathbf{H}$, so Gram aggregation is insensitive to

frame order. This property is useful for effect textures whose perceptual identity depends on stationary co-activation patterns rather than exact temporal alignment. Figure 2 visualizes the resulting representation at two levels: an example Gram matrix that exposes the co-activation structure used by TRR, and a t-SNE diagnostic comparing TRR with mean-pooled Wav2Vec2 embeddings. These plots are interpretive diagnostics; the quantitative claim remains anchored to the held-out retrieval and projection metrics.

C. Protocol-B: Exemplar-Preserving Projection

Top-1 retrieval returns an executable preset, but it is unnecessarily brittle when several nearby presets form a more parameter-transferable neighborhood. We therefore introduce **Exemplar-Preserving Projection (EPR-K5)** as the projection module. Given a query q , a retrieval prior returns a top- K executable neighborhood

$$\mathcal{N}_K(q) = \{(\theta_i, s_i)\}_{i=1}^K, \quad (5)$$

where θ_i is a stored executable preset and s_i is the retrieval similarity score. We fix $K = 5$ and temperature $\tau = 0.05$ for all methods and do not tune these hyperparameters on test labels. The exemplar weights are

$$w_i = \frac{\exp(s_i/\tau)}{\sum_{j=1}^K \exp(s_j/\tau)}. \quad (6)$$

For continuous numeric leaves, EPR computes a weighted average and clips each leaf to the observed range in the knowledge base:

$$\tilde{\theta}_{\text{cont}} = \Pi_\Theta \left(\sum_{i=1}^K w_i \theta_{i,\text{cont}} \right). \quad (7)$$

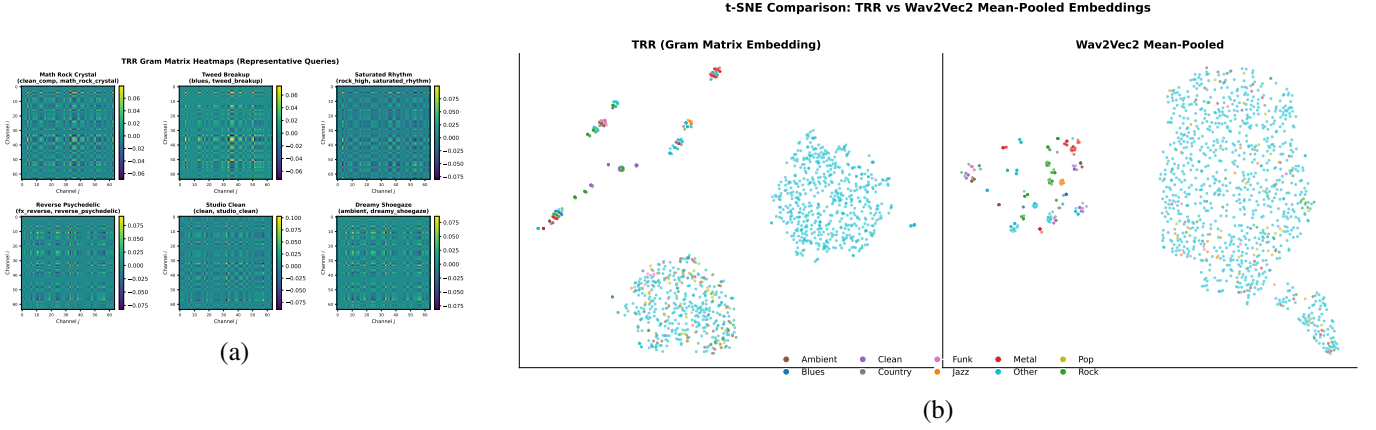


Fig. 2. TRR representation diagnostics. (a) Example second-order Gram matrix. (b) t-SNE comparison between TRR and mean-pooled Wav2Vec2 embeddings on the benchmark presets. The figure is used for qualitative inspection rather than as standalone evidence of parameter transferability.

For binary module switches, EPR uses a weighted vote over the top- K candidates and activates a switch when its weighted support exceeds 0.5:

$$\tilde{\theta}_{\text{on}} = \mathbb{I} \left[\sum_{i=1}^K w_i \theta_{i,\text{on}} > 0.5 \right]. \quad (8)$$

Missing keys are not artificially generated; as in the evaluator, missing numeric leaves enter the union-key metric as zeros. This keeps the projection comparable with the retrieval setting and avoids silently adding parameters that no retrieved exemplar supplies. We report provenance with the largest exemplar contribution $\max_i w_i$ and the inverse-Herfindahl effective exemplar count

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^K w_i^2}. \quad (9)$$

EPR-K5 produces a new parameter vector that is not itself a stored preset: it is a softmax-weighted blend of retrieved exemplars followed by range and switch projection. Its output remains executable because every parameter value is either clipped to the observed range of retrieved neighbors or decided by a weighted vote over them. The neighborhood provenance (N_{eff} and $\max_i w_i$) documents which retrieved presets contributed, but the final vector is a synthetic projection rather than a single retrieved exemplar.

D. Complexity and Latency

TRR encoding includes Wav2Vec2 feature extraction, then $O(D^2 T_{\text{frm}})$ Gram construction after projection with $D = 64$, followed by top- K nearest-neighbor search. Cached retrieval scoring is lightweight in the local artifact, but uncached live reference-audio latency is not claimed in the main paper because the reviewed artifact does not expose a complete live Wav2Vec2-forward profiling path.

V. EXPERIMENTS

We validate the proposed framework by answering one core research question: does modeling texture with mid-level deep-feature statistics (TRR) improve parameter inference compared to first-order baselines?

A. Experimental Setup

a) Dataset and Knowledge Base.: The benchmark corpus contains 1267 paired records (audio–text–parameter triplets) collected from guitar effect presets. A post-hoc audit of the original song-name-based held-out list (211 test queries, 1056 knowledge-base items) revealed 16 cross-split shared resolved audio paths and 48 near-duplicate cross-split pairs under a cheap fingerprint. To remove exact audio reuse from the main objective benchmark, we regroup records by resolved local audio path before sampling the held-out set. The resulting stricter held-out split contains 204 test queries and 1063 knowledge-base items, with zero exact shared resolved audio paths across split under the current audit. We retain this as a pilot benchmark because cheap near-duplicate pairs remain and the corpus is still guitar-only.

b) Provenance notes.: The dataset JSON exposes song name, style tags, feature tags, parameters, and stored vectors, but it does not include a separate auditable text-source field. We therefore treat the style and feature fields as the only explicit text-side provenance available in-repo, and use the stored parameters as the operational ground truth.

c) Scope and protocol.: Unless otherwise stated, objective results in this section report **top-1 retrieval** and **EPR-K5 projection** on the same 204-query split and the same 1,063-record retrieval database. Degradation-robustness, listening tests, and latency measurements are reported on their own experimental pools and are labeled explicitly.

d) Resources and reproducibility.: The supplementary package contains preprocessing details, embedding computation steps, retrieval-index construction, prompt templates, evaluation scripts, and split specifications for all reported tables. The artifact is anonymized for review and makes each reported table traceable to a script-level evidence path.

e) Baselines.: We compare our proposed **TRR** (Section IV-B) against locally auditable retrieval baselines. **Wav2Vec** uses mean-pooled Wav2Vec2 activations. **FeatureNN** uses the released benchmark feature vectors and should therefore be interpreted as a fixed dense-retrieval reference. **CLAP** uses the stored CLAP vectors available in the local JSON artifact and is reported with its actual coverage. **PaSST** is regenerated in the

E9 derived artifact with full top-1 retrieval coverage and serves as the full-coverage modern audio-encoder baseline. PANNs remains excluded from unified tables due to model weight download constraints; we do not claim “best full-coverage” unconditionally.

B. Metrics

We evaluate alignment in parameter space Θ using operational metrics computed on flattened numeric parameter leaves. Concretely, given a nested parameter dictionary, we recursively flatten numeric leaves into a sparse vector indexed by dot-separated keys (e.g., `DelayOn.Mix`) and ignore non-numeric leaves (e.g., strings). Unless otherwise stated, we form the union of keys between prediction and ground truth, and treat missing numeric keys as zeros for metric computation. The unified main table reports normalized L2, normalized Acc@0.1, Recall, Cosine, value-aware SwitchF1, coverage, and provenance. The earlier key-presence Module Jaccard is retained only as an audit diagnostic because it can reward methods that emit a complete schema even when switch values are not aligned. Raw L2 magnitudes can be dominated by a small number of large-scale parameters and are therefore retained only in protocol-specific audit artifacts. Normalized L2 uses plugin-defined min-max ranges for comparability, but this is still an engineering normalization rather than a perceptual scale: logarithmic, dB, and nonlinear control laws can make equal normalized parameter differences perceptually unequal.

a) *L2 Error (RMSE).*: Given flattened vectors $v(\theta_{\text{gt}})$ and $v(\tilde{\theta})$ over the union key set of size d , we report the root mean squared error

$$L2 = \sqrt{\frac{1}{d} \sum_{i=1}^d \left(v(\theta_{\text{gt}})^{(i)} - v(\tilde{\theta})^{(i)} \right)^2}. \quad (10)$$

b) *Normalized Accuracy@0.1.*: For the unified tables, we report the fraction of normalized flattened numeric parameters whose absolute error is within a fixed tolerance $\tau = 0.1$:

$$\text{Norm.Acc@0.1} = \frac{1}{d} \sum_{i=1}^d \mathbb{I} \left(\left| v(\theta_{\text{gt}})^{(i)} - v(\tilde{\theta})^{(i)} \right| \leq 0.1 \right). \quad (11)$$

c) *Recall (active numeric parameters).*: We define a ground-truth numeric parameter as *active* if $|v(\theta_{\text{gt}})^{(i)}| > 0.05$. Recall is the fraction of active ground-truth dimensions whose predicted value falls within the same tolerance $\tau = 0.1$.

d) *Cosine Similarity.*: We compute cosine similarity between flattened vectors over the union key set:

$$\text{Cosine} = \frac{v(\theta_{\text{gt}}) \cdot v(\tilde{\theta})}{\|v(\theta_{\text{gt}})\|_2 \|v(\tilde{\theta})\|_2}. \quad (12)$$

e) *Switch-State Metrics.*: We report value-aware active-switch precision, recall, and F1 rather than using key presence alone. For each switch root, the evaluator infers the active state from the numeric `On/Off` leaves and compares the predicted active-switch set S_{pred} against the ground-truth set S_{gt} :

$$\text{SwitchF1} = \frac{2|S_{\text{gt}} \cap S_{\text{pred}}|}{|S_{\text{gt}}| + |S_{\text{pred}}|}. \quad (13)$$

The legacy Module Jaccard over keys ending with `On` is still written to the audit JSON, but it is not used as a main evidence column.

f) *Parameter-Neighborhood Recall.*: Because the task is executable-neighborhood retrieval rather than only top-1 nearest-preset selection, we also report parameter-neighborhood recall:

$$\text{PNR@K} = \mathbb{I} \left[\min_{i \in \text{TopK}(q)} d_{\theta}(\theta_i, \theta_q) < \tau_p \right], \quad (14)$$

where d_{θ} is normalized parameter RMSE. The primary table uses $\tau_p = 0.10$ and the supplementary audit reports $\tau_p \in \{0.05, 0.10, 0.15\}$ sensitivity.

C. Protocol-A: Comparative Retrieval Evaluation

Purpose. Benchmark top-1 executable retrieval and EPR-K5 exemplar-preserving projection under one evaluator, one split, and one normalization path.

Protocol. Protocol-A uses top-1 retrieval on a deterministic 204-query test split and a 1,063-record retrieval database. Protocol-B applies the EPR-K5 projection defined in Section IV-C to the same retrieved neighborhoods. Both protocols use the same split, evaluator, normalized parameter distance, missing-key convention, and coverage accounting. A method is reported with its actual number of evaluable queries rather than with imputed missing vectors.

The paper-facing evidence separates quantitative artifacts from illustrations. Protocol-A/B metrics, robustness curves, and EPR sensitivity are tied to exported CSV/JSON artifacts with fixed split files and coverage accounting; mechanism plots, error diagnostics, listening summaries, and framework diagrams are used only as diagnostics or non-result illustrations. The detailed provenance audit is retained in the source package rather than expanded in the main text.

Table I reports the unified comparison. PaSST is regenerated in the E9 derived artifact with full top-1 retrieval coverage and is included in both the retrieval and projection comparisons. PANNs remains excluded due to model weight download constraints; we therefore do not claim a fair PANNs EPR-K5 baseline.

*Direct-regression boundary: MLP matches TRR EPR-K5 on Norm.Acc@0.1 (0.7894 vs. 0.7908; $\Delta = 0.0014$) but lacks retrieved-neighborhood provenance and is substantially weaker on Recall (0.3468 vs. 0.7146) and Cosine (0.5941 vs. 0.9207). Retained to show that the TRR advantage is not primarily on tolerance accuracy but on recall, cosine alignment, and provenance. †The MLP regressor predicts continuous numeric leaves and does not emit discrete switch states; SwitchF1 is therefore undefined. See Supplementary Section S11.

‡Dimensionality control: a fixed random projection of the 768-D Wav2Vec mean-pooled vector to 4096-D. Only Norm.L2 is reported because the control is intended solely to rule out the “higher dimensionality” explanation for TRR’s advantage.

Retrieval results. Under the unified exporter, TRR top-1 is the strongest full-coverage top-1 retrieval row on normalized distance, active-parameter recall, cosine, and switch F1. Bootstrap 95% CIs show that TRR’s Norm.L2 advantage over all

TABLE I

UNIFIED RETRIEVAL AND PROJECTION COMPARISON ON THE SHARED 204-QUERY BENCHMARK. TOP-1 ROWS ARE EXECUTABLE RETRIEVAL; EPR-K5 ROWS APPLY NEIGHBORHOOD PROJECTION TO THE SAME RETRIEVAL PRIOR. THE MLP ROW* IS A DIRECT-REGRESSION BOUNDARY WITHOUT RETRIEVED-NEIGHBORHOOD PROVENANCE, INCLUDED TO DELINEATE THE RETRIEVAL-VS-REGRESSION TRADE-OFF. THE DIMENSIONALITY CONTROL[‡] RULES OUT THE “HIGHER DIMENSIONALITY” EXPLANATION FOR TRR’S ADVANTAGE. COVERAGE IS $n/204$. NORM.ACC@0.1 IS COMPUTED BY THE NORMALIZED EVALUATOR. SWITCHF1 IS THE VALUE-AWARE ACTIVE SWITCH F1; THE LEGACY KEY-PRESENCE MODULE JACCARD IS RETAINED ONLY IN THE AUDIT. LOWER NORM.L2 IS BETTER; HIGHER IS BETTER FOR THE OTHER METRICS. BOOTSTRAP 95% CIs FOR NORM.L2 ARE SHOWN IN PARENTHESES (10,000 RESAMPLES OF QUERIES).

Method	Cond.	n	Cov.	Norm.L2 ↓	95% CI	Acc@0.1 ↑	Rec. ↑	Cos ↑	Sw.F1 ↑	Prov.
Wav2Vec top-1	Top-1	204	1.000	0.1997	[0.1846, 0.2148]	0.7085	0.6228	0.8349	0.7692	yes
Wav2Vec EPR-K5	EPR-K5	204	1.000	0.1838	[0.1713, 0.1966]	0.7030	0.6194	0.8622	0.7886	EPR
FeatureNN top-1	Top-1	204	1.000	0.2002	[0.1851, 0.2151]	0.7055	0.6208	0.8435	0.7898	yes
FeatureNN EPR-K5	EPR-K5	204	1.000	0.1878	[0.1764, 0.1993]	0.6864	0.6135	0.8640	0.7970	EPR
CLAP top-1	Top-1	173	0.848	0.1896	[0.1727, 0.2068]	0.7204	0.6361	0.8538	0.8125	yes
CLAP EPR-K5	EPR-K5	173	0.848	0.1642	[0.1519, 0.1765]	0.7286	0.6447	0.8967	0.8553	EPR
PaSST top-1	Top-1	204	1.000	0.1960	[0.1808, 0.2111]	0.7063	0.6256	0.8463	0.7970	yes
PaSST EPR-K5	EPR-K5	204	1.000	0.1808	[0.1692, 0.1926]	0.6965	0.6170	0.8723	0.8123	EPR
TRR top-1	Top-1	204	1.000	0.1454	[0.1301, 0.1606]	0.7863	0.7045	0.9010	0.8630	yes
TRR EPR-K5	EPR-K5	204	1.000	0.1334	[0.1211, 0.1462]	0.7908	0.7146	0.9207	0.8824	EPR
MLP + RangeProjection*	Direct	204	1.000	0.1603	[0.1530, 0.1678]	0.7894	0.3468	0.5941	— [†]	no
Wav2Vec 4096-D [‡]	Control	204	1.000	0.1999	—	—	—	—	—	no

full-coverage baselines is stable (no overlap with Wav2Vec, FeatureNN, or PaSST intervals). Pairwise Wilcoxon signed-rank tests against all full-coverage baselines yield $p < 0.001$ with medium to large effect sizes ($r > 0.35$); all comparisons remain significant after Holm-Bonferroni correction (largest corrected $p = 1.63 \times 10^{-5}$).

The dimensionality-matched control in Table I ([‡]) projects the 768-D Wav2Vec mean-pooled vector to 4096-D via a fixed random projection, achieving Norm.L2 = 0.1999—statistically tied with the original 768-D baseline (0.1997) and far above TRR (0.1454)—thereby ruling out the “higher dimensionality” alternative explanation. The range-projected MLP remains useful as a direct-regression baseline: it matches selected tolerance scores, but it has no retrieved-neighborhood provenance and is substantially weaker on Recall and Cosine. Its legacy Module Jaccard was 1.0000 because that audit metric detects top-level On keys rather than value-aware switch states; we therefore do not use it as a main evidence column.

Projection results. Applying EPR-K5 to TRR further reduces Norm.L2 from 0.1454 to 0.1334 while retaining neighborhood provenance.

Under the unified comparison, TRR is the strongest full-coverage retrieval prior among evaluated baselines on this benchmark: its texture-aware mid-level representations better align with executable parameter neighborhoods than the available first-order and dense-retrieval baselines. PaSST provides a full-coverage modern audio-encoder baseline; PANNs is excluded due to model weight download constraints documented in the artifact.

D. Mechanism and Error Diagnostics

Beyond the leaderboard, we include mechanism-level diagnostics to clarify what part of TRR is being supported. Figure 3 checks whether mid-level Wav2Vec2 activations are a plausible operating region for TRR, while Figure 4 shows where normalized parameter error concentrates across the effect-chain leaves. These diagnostics should not be read as independent proof that the selected layer set is globally

optimal; rather, they document that the reported configuration is a defensible operating point under the current benchmark.

The diagnostics are consistent with the bounded mechanism interpretation used in the supplementary material. Same-backbone Gram aggregation improves over mean pooling in the archived and local mechanism checks, but the gain is partly entangled with mid-level layer choice and projection dimension. Controlled ablation shows that Gram statistics improve normalized L2 by 0.0125 at layer 5 alone (Wilcoxon $p = 0.586$) but only 0.0007 at multi-layer $\{4, 5, 6\}$ ($p = 0.767$); neither paired test reaches significance, yet the effect-size difference between layer-5 and multi-layer confirms that the advantage is coupled with layer selection. For this reason, the paper claims TRR as a useful texture-aware retrieval prior, not as a globally optimized audio representation family. The per-parameter view also makes clear that a lower aggregate normalized distance does not eliminate localized weaknesses: switch-like structures, sparse effect leaves, and parameters with nonlinear perceptual scales still require separate validity and perceptual audits.

E. Robustness and Boundary Diagnostics

We include compact robustness diagnostics in the main paper and move full sweeps, confidence intervals, EPR sensitivity, validity repair rates, and listening-test tables to the supplementary material. First, CLAP has partial coverage in the unified benchmark. On the 173-query shared subset where every retrieval prior has query and candidate vectors, TRR remains best on top-1 normalized distance (0.1487), PNR@5 (0.8150), and EPR-K5 normalized distance (0.1328). Second, formal parameter-space near-duplicate filtering at three thresholds (0.02, 0.05, 0.10) confirms that TRR maintains its advantage at strict thresholds: at threshold 0.02 (KB 1063→84), TRR Norm.L2= 0.213 vs. Wav2Vec= 0.237; at threshold 0.05 (KB→63), TRR= 0.201 vs. Wav2Vec= 0.231. At the most aggressive threshold (0.10, KB→36), the advantage shrinks but persists (TRR= 0.223 vs. Wav2Vec= 0.229). Third, parameter-cluster hard splits provide a stronger stress

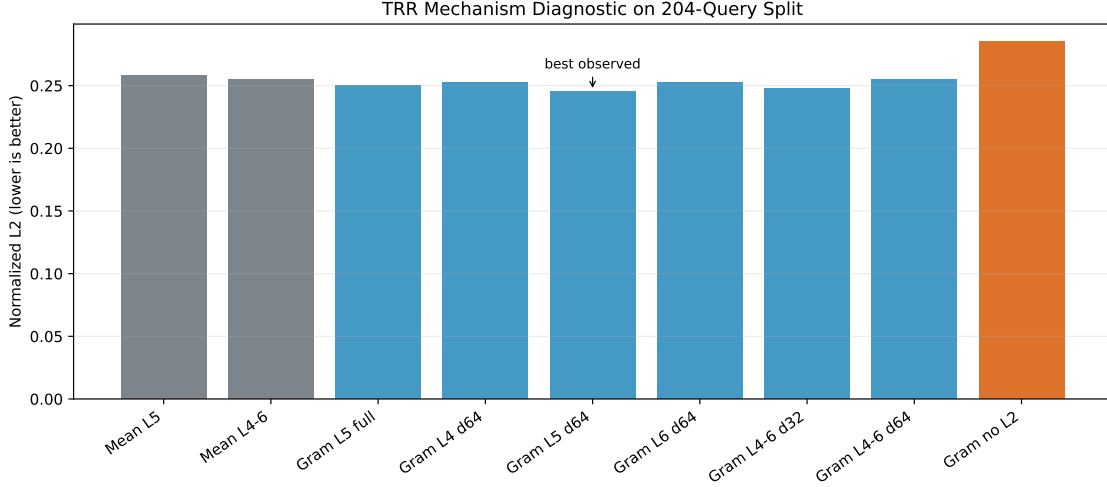


Fig. 3. TRR mechanism and layer-selection diagnostic (204-query held-out split, normalized L2). The plot provides bounded evidence about which Wav2Vec2 layer region and aggregation choice give the most useful texture representation under the current benchmark.

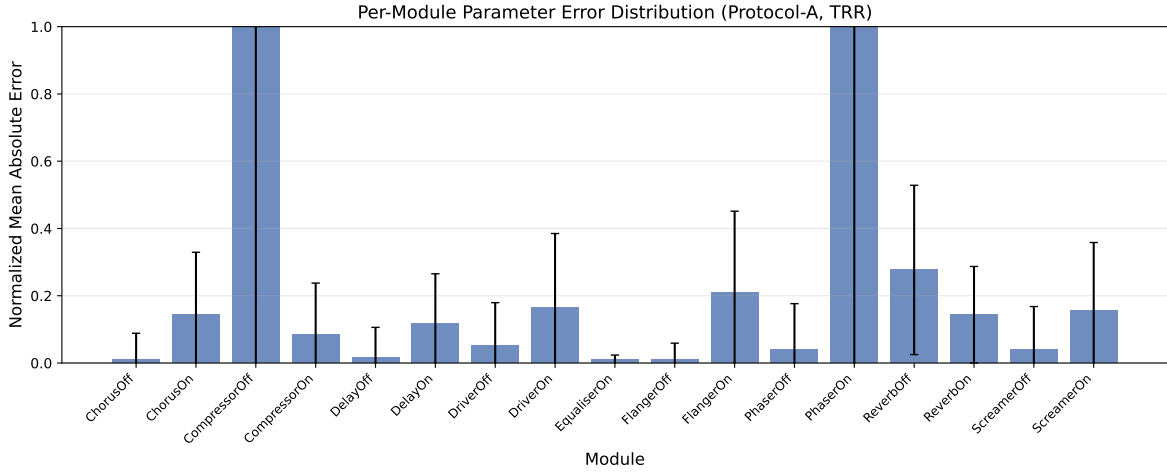


Fig. 4. Per-parameter error distribution diagnostic (204-query held-out split, normalized L2). The figure complements the aggregate retrieval and projection metrics by showing where normalized parameter errors concentrate across effect-chain leaves.

TABLE II
COMPACT ROBUSTNESS DIAGNOSTICS. NEAR-DUPLICATE ROWS REPORT TRR NORMALIZED L2 AFTER REMOVING CANDIDATES WITHIN THE THRESHOLD OF ANY QUERY. HARD-SPLIT ROWS REPORT THE BEST FULL-COVERAGE TRR NORMALIZED L2 UNDER PARAMETER-CLUSTER EXCLUSION. FULL BASELINE SWEEPS AND CONFIDENCE INTERVALS ARE PROVIDED IN THE SUPPLEMENTARY MATERIAL.

Diagnostic	Setting	KB / n_q	TRR Norm.L2
No filter	threshold 0.000	1063 / 204	0.1454
Near-duplicate filter	threshold 0.010	342 / 204	0.1968
Near-duplicate filter	threshold 0.020	151 / 204	0.2510
Hard split	$\tau_{\text{cluster}} = 0.01$	278 / 50	0.2065
Hard split	$\tau_{\text{cluster}} = 0.02$	142 / 46	0.2424
Hard split	$\tau_{\text{cluster}} = 0.05$	117 / 38	0.2668

test in which query clusters are removed from the database. TRR remains the lowest-error full-coverage row at all tested thresholds; CLAP is competitive on its covered subset but has fewer evaluable hard queries.

Figure 5 plots the same robustness evidence as curves. The near-duplicate panel shows that TRR degrades as candidate density is reduced, but remains the strongest full-coverage row through the reported thresholds. The hard-split panel shows the expected increase in error when parameter clusters are excluded from the database; this is a stress-test result rather than a production setting. The figure is generated from the existing E9 robustness artifacts and is included to make the degradation trend inspectable in the main paper.

The direct-regression boundary and projection diagnostics further delimit the claim. The range-projected MLP row in Table I nearly matches TRR EPR-K5 on normalized tolerance, but it is weaker on normalized distance, recall, and cosine similarity and has no retrieved-neighborhood provenance. TRR top-K retrieval reaches $\text{PNR@5}=0.7500$ on the 204-query held-out split, and TRR EPR-K5 is the primary hybrid result with $N_{\text{eff}} = 4.2129$ and $\max_i w_i = 0.3438$. Figure 6 is retained as main-method evidence because it justifies the EPR-K5 operating point: increasing K improves

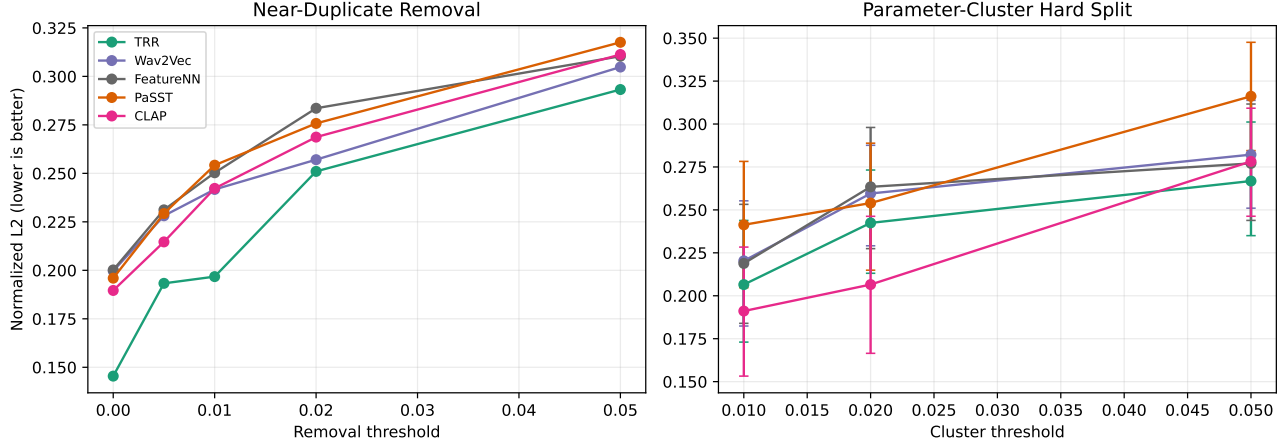


Fig. 5. Robustness result curves from the existing E9 artifacts. Left: near-duplicate removal sensitivity. Right: parameter-cluster hard-split sweep with bootstrap confidence intervals where available. Lower normalized L2 is better.

normalized distance, while also spreading provenance over more exemplars. We therefore report K5 as the main hybrid operating point because it improves over top-1 while retaining a compact neighborhood. EPR-K5 hyperparameters ($K = 5$, $\tau = 0.05$) are not fine-tuned: a full grid search over $K \in \{3, 5, 10, 20\} \times \tau \in \{0.01, 0.05, 0.10, 0.20, 0.50\}$ shows Acc@0.1 varies by $< 1.2\%$ (0.7809–0.7930), confirming robustness to hyperparameter choice. A module-aware reranking diagnostic is negative in this run, increasing Norm.L2 by 0.0024 and decreasing Recall and Cosine; it is therefore not reported as a gain.

F. Degraded-Input and Listening Diagnostics

We also surface two diagnostic groups that were previously only summarized textually. First, degraded-input fusion re-encodes waveform degradations (AWGN, MP3, reverberation, and truncation) before retrieval. The best adaptive fusion setting obtains $\text{L2}=20.1491$, $\text{Acc@0.1}=0.6121$, $\text{MRR}=0.5654$, and $\text{NDCG@5}=0.5566$, but its margin over strong fixed text-weighted fusion is small. We therefore treat adaptive fusion as a supplementary robustness-boundary diagnostic rather than as evidence for the core TRR/EPR effectiveness claim; the weight-distribution and failure-case plots are reported in the supplementary material.

Second, a 26-participant multiple-stimulus listening study provides perceptual context. Figure 7 summarizes the manual-baseline comparison trials; the remaining listening distributions are reported in the supplementary material. The observed distributions support a cautious interpretation: the TRR-based system is competitive in the tested listening setup, but the study is not a standard MUSHRA protocol and does not validate the objective metrics as human-rating proxies. Because it lacks a standard low-quality anchor, device metadata, participant-expertise labels, and a documented manual-tuning time budget, we treat it as supporting context rather than a primary validation of the retrieval or projection protocols. A metric-perception alignment diagnostic is also negative under the available alias mapping. Together, these results reinforce that the main claim is parameter-space transferability under

audited retrieval protocols, not a validated predictor of human listening scores.

VI. DISCUSSION

A. Scope of the Current Evidence

The core quantitative evidence is the unified retrieval and projection benchmark on a 204-query guitar-only split, supplemented by near-duplicate sensitivity, CLAP-covered shared-subset analysis, EPR sensitivity, EPR validity/repair auditing, mechanism ablations, and the supervised direct-regression boundary. Degradation-robustness and listening/local-vLLM diagnostics provide additional stress-test and perceptual characterization. All claims about texture-aware retrieval are bounded to this benchmark; generalization to other instruments, effect families, or production stages is hypothesized but not validated. Personalization and memory components are outside the paper’s validated claims because they are not evaluated under the same held-out protocol.

B. Implications for Audio Effect Control

The current results suggest a practical pattern for retrieval-based audio control: use retrieval over real presets to provide feasibility constraints and stylistic priors, and then expose the retrieved configuration for direct user editing. In the top-1 retrieval setting, TRR achieves the strongest parameter alignment among the full-coverage rows while preserving editability and traceability in DAW workflows. The projection setting sharpens the boundary: MLP+RangeProjection is a direct-regression baseline, while EPR-K5 is the strongest hybrid result when executable-neighborhood provenance is part of the target. Together, these results support TRR as a retrieval prior with measurable parameter-transferability benefits on this benchmark.

a) *Retrieval versus regression.*: The MLP boundary clarifies the fundamental trade-off between retrieval-based and regression-based approaches. On Norm.Acc@0.1 , MLP (0.7894) is effectively tied with TRR EPR-K5 (0.7908); the gap of 0.0014 is well within sampling variability. However, on

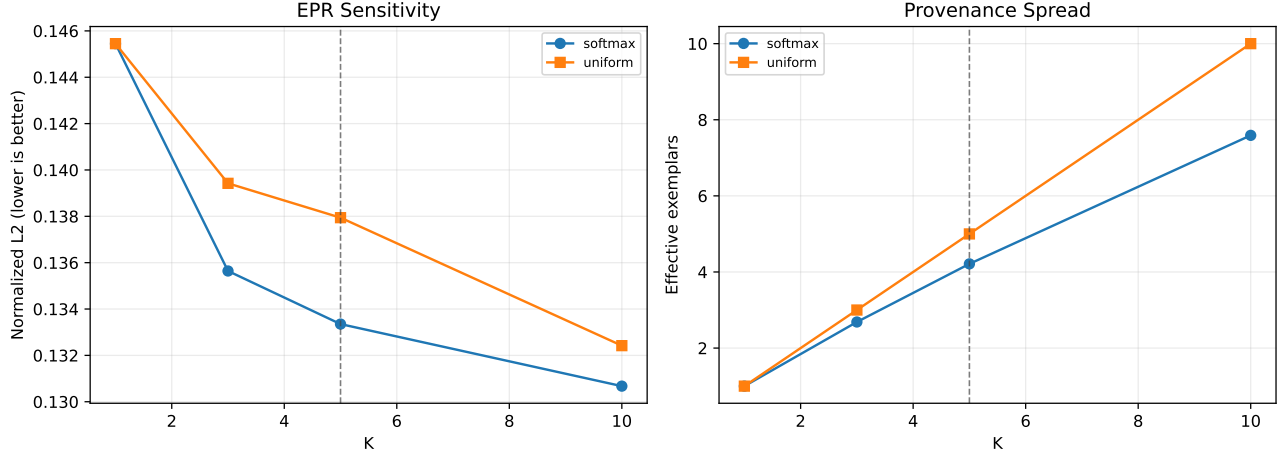


Fig. 6. EPR sensitivity from the existing E9 artifact. Left: normalized L2 as a function of K , weighting, and softmax temperature. Right: effective exemplar count, showing the provenance-spread trade-off. The dashed line marks the $K5$ operating point used in the main table.

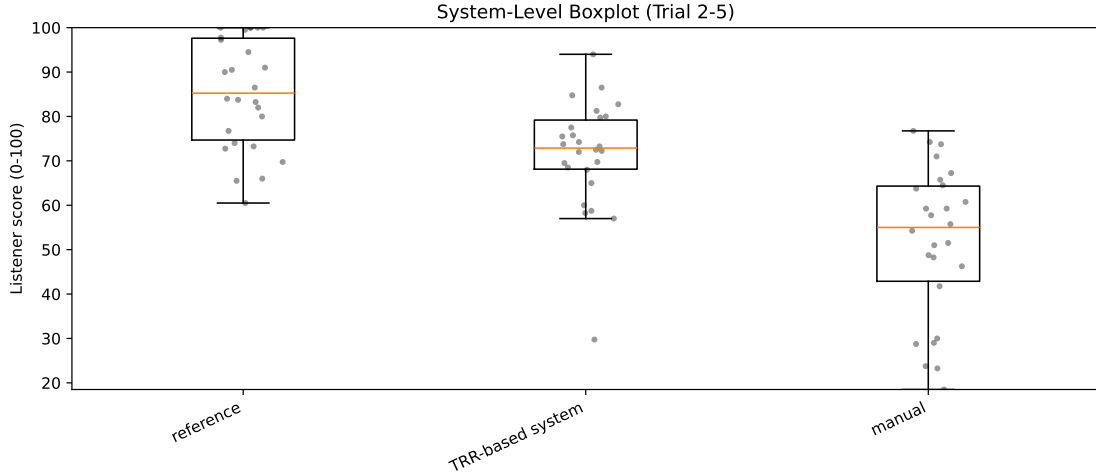


Fig. 7. Representative listening-study distribution for Trials 2–5, comparing the TRR-based system with a manual baseline. The figure is supporting perceptual context rather than primary validation.

Recall the gap is large (0.3468 vs. 0.7146), indicating that the MLP regressor fails to activate the same set of parameters as ground truth even when its numeric values are within tolerance on the dimensions it does predict. Similarly, Cosine similarity (0.5941 vs. 0.9207) shows that the MLP’s predicted parameter vector is structurally misaligned with ground truth. The retrieval approach succeeds because it anchors predictions to real preset exemplars: every retrieved parameter is drawn from an executable neighborhood, so the output inherits the activation patterns and correlations of real presets. Regression, by contrast, produces point estimates that may be numerically close on individual parameters but lack the global structure of valid presets. This trade-off—numerical precision versus provenance and structural alignment—is the central design rationale for TRR: the system is not primarily motivated by tolerance accuracy, but by the ability to produce outputs that are structurally coherent, editable, and traceable to specific exemplars.

b) Stakeholder considerations.: Different user groups may experience the system differently. Amateur guitarists may value one-click preset retrieval and simple top-1 selection, whereas professional session musicians and sound designers may prefer browsing a top- K gallery or inspecting the provenance weights of an EPR- $K5$ blend. Mixing engineers care about how a retrieved guitar tone sits in a full mix, a perspective not captured by isolated parameter-space metrics. Plugin developers must weigh the storage and latency costs of a retrieval index against the benefits of semantic search. The current evaluation does not distinguish these stakeholders; future HCI work should conduct task-based studies with representative user groups to assess whether retrieval-based control improves workflow efficiency and creative satisfaction.

C. Revisiting the Texture Hypothesis

In the retrieval benchmark, TRR improves parameter-space alignment over the mean-pooled Wav2Vec baseline. This comparison controls for the acoustic backbone and changes

the aggregation operator from first-order pooling to second-order Gram statistics. The supplementary mechanism ablation reinforces this interpretation: on the same 204-query split, a same-layer Gram representation improves normalized L2 over mean pooling, the full unprojected Gram boundary also improves over mean pooling, and removing L2 normalization degrades retrieval quality. It therefore supports the hypothesis that correlation structure can be useful as a texture descriptor for retrieval.

We note an important caveat: the backbone-controlled comparison changes both the aggregation operator (mean pooling $\rightarrow$ Gram matrix) and the feature processing pipeline (single-layer final representation $\rightarrow$ multi-layer mid-level projection). The supplementary ablation (Table S5) now includes a same-layers mean-pooling baseline: mean-pooling across layers 4–6 improves only marginally over mean-pooling layer 5 alone (0.2552 vs. 0.2582), and multi-layer Gram with $d = 64$ improves only marginally over multi-layer mean pooling (0.2546 vs. 0.2552). The clearest Gram advantage appears on single-layer 5 (0.2457 vs. 0.2582). The gain is therefore consistent with the texture hypothesis but not experimentally isolated from mid-level layer selection: the advantage could reflect the suitability of layer 5 for this benchmark as much as the superiority of Gram statistics over mean pooling. The supplementary mechanism ablation shows that the Gram advantage at layer 5 (+0.0125) is substantially larger than at multi-layer (+0.0007), and neither reaches statistical significance (Wilcoxon $p > 0.05$). We therefore frame TRR as a texture-aware retrieval prior using mid-level features: the Gram matrix is an effective aggregation operator, but the primary driver of performance is the choice of mid-level layer 5.

The derived per-query analysis further characterizes where TRR matters most: texture-aware correlation structure is especially beneficial when coarse semantic similarity alone is insufficient and retrieval depends on texture-specific co-activation patterns. The ablation shows that layer selection and projection dimension are not cosmetic details; the reported implementation is a practical operating point. We interpret TRR as a retrieval prior for texture-sensitive control that complements first-order semantic representations.

An open direction is whether analogous second-order representations transfer to other audio-effect settings beyond the current benchmark, including additional instruments, effect families, and mix-stage tasks. Future work can validate this through controlled cross-domain protocols and human preference judgments.

D. Ethical Considerations

The current retrieval-grounded system is intended to augment user creativity rather than replace it. Risks include homogenization toward common exemplar neighborhoods and over-reliance on suggested settings. We examined the genre distribution of retrieved presets and found no evidence of extreme bias toward a single dominant style, but the corpus itself is guitar-centric and may underrepresent niche genres. In future deployments, the interface should emphasize transparency (e.g., showing retrieved references) and user

agency (e.g., direct editability and opt-out of retrieval). More generally, any model or retrieval corpus built from existing recordings or presets should consider copyright, licensing, and data provenance constraints.

VII. LIMITATIONS

The evidence is bounded in four ways. First, the objective benchmark is a 204-query, guitar-only split with one effect-chain topology; resolved-audio grouping removes exact cross-split reuse, but near-duplicate risk, corpus density, and cross-domain generalization remain open. Second, normalized parameter metrics are engineering proxies rather than perceptual scales, and the listening diagnostic is not a standard MUSHRA-style, query-aligned user study; we therefore do not claim that parameter-space gains directly imply perceived quality gains or workflow improvement. Third, the baseline claim is conditional on the evaluated methods: CLAP, PaSST, and same-backbone ablations strengthen the comparison, whereas PANNs remains excluded because its model weights were not available in the artifact environment. Fourth, EPR-K5 is a projected blend of retrieved exemplars rather than pure retrieval, so its range/switch repair step is part of the deployed method. Personalization, memory, adaptive fusion, uncached live-reference latency, and out-of-distribution database expansion are outside the validated scope.

These limitations define the scope of our current claims: TRR is supported as a useful texture-aware retrieval prior for editable preset selection on this benchmark, while broader perceptual, user-centered, and deployment claims require separate validation.

VIII. CONCLUSION

We presented parameter-transferability-aware executable neighborhood retrieval for editable guitar-effect preset selection and identified Texture Resonance Retrieval (TRR) as the strongest full-coverage retrieval representation among evaluated baselines in the unified local comparison. Across the unified retrieval and projection metrics (Table I), TRR improves parameter alignment over Wav2Vec, FeatureNN, and PaSST, and EPR-K5 improves the TRR retrieval prior while retaining neighborhood provenance. These results support the utility of texture-aware retrieval as a prior for executable preset selection on this benchmark.

Taken together, the retrieval and projection evidence supports a practically relevant conclusion: retrieval over executable presets provides editable starting points for DAW workflows, mid-level texture-aware feature representations improve retrieval over selected first-order baselines on the audited guitar-effect preset benchmark, and exemplar-preserving projection improves numeric alignment while retaining provenance after explicit validity projection. The degradation, listening, local-LLM, module-aware reranking, PANNs model-weight download audit, and supervised-regression diagnostics provide complementary boundary characterization. Our claims remain conditional on the evaluated baselines and do not preclude untested baselines (e.g., PANNs) from achieving comparable results.

Looking forward, the most important next steps are to evaluate executable direct parameter-regression and reranking baselines, expand the degradation and listening protocols, and test whether the same retrieval and fusion advantages transfer beyond guitar effects to broader production parameter spaces such as mixing and mastering workflows.

Overall, these results support texture-aware retrieval as a useful prior for editable guitar-effect preset selection on the current benchmark. Natural next directions include evaluating the framework on broader instrument classes and effect families, integrating personalization and memory components, and validating the approach under more diverse production workflows.

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